

Natural Language Processing

- NLP is a field that combines Computer Science and Language Studies.
- It helps computers understand, process and create human language in a way that makes sense and is useful.
- Field of AI focused on the interaction b/w computers and human language.

NLP is used by many applications that use language, such as text translation, voice recognition, text summarization and chatbots.

- Chatbots & Virtual assistants (e.g. Siri, Alexa)
- Language Translation (Google Translate) Encoder Decoder
↑ ↓
English Hindi
- Search Engines
- Spam Detection
- Social media Monitoring
- Legal and medical document Processing.

Goals of NLP

- Understanding language (Syntax, Semantics, Pragmatics)
- Generating language (Translation, Summarization, Dialogue)
- Processing and analyzing large volume of natural language data.

Key NLP Tasks

- Tokenization → Splitting text into words, phrases or symbols
- Part of speech Tagging → Identifying nouns, verbs, adjectives, etc.
- NER (Named Entity Recognition) → Detecting names of people, places, dates etc.
- Sentiment Analysis - Determining the sentiment (positive, negative, neutral)

Machine Translation → Translating text from one language to another

Text Summarization → Creating concise summaries of longer texts.

Question Answering → Answering questions from a passage or database.

Speech Recognition → Converting spoken language into written text.

Core Concepts

- Syntax (grammar, structure of sentences)
- Semantics (Meaning of words / sentences)
- Pragmatics (Contextual language use)
- Corpus - (A large collection of text used to train NLP models)

Common Tools & Libraries

- NLTK
- spaCy
- TextBlob
- Stanford NLP
- Hugging face Transformers (BERT, GPT, etc).

NLP Techniques

① Text Processing & Preprocessing

Tokenization (Dividing)

Stemming and Lemmatization (root forms)

Stopword Removal (Removing common words "is", "and")

Text Normalization (Standardizing text)

21/11/25. Levels of NLP 1) Phonology / Phonetics

- ② Morphology → study of structure of words
cat /kæt/ unhappiness
un + happy + ness
- ③ Lexical Analysis → To learn grammatical structure
- ④ Syntactic Analysis → correctness of grammar Bank, river, bank

- ⑤ Semantic Analysis : I am going to bank, to wind down for fishing.
- ⑥ Pragmatics : can you pass me a ham (try to understand the meaning of the sentence).
- ⑦ Syntax and Parsing

① Part of speech (Pos) Tagging

→ Assigning parts of speech to each word in a sentence. Such as nouns, verbs, adjectives and adverbs.

- ① Noun : Person place, things
- ② Verb : action word.
- ③ Pronoun : Substitute of a noun
(He, she, our, you, my)
- ④ Adjective : Describes the noun. Adds meaning to it
- ⑤ Adverb : Describes verb
- ⑥ Interjection : wow, hey! Alas!
- ⑦ Conjunction : but, and, or, either or, neither nor.
(connecting words)
- ⑧ Preposition : Links a noun to another word.
in, on, at, after, against, along, among, around, because of, before, behind, below, beneath, besides, between.
- ⑨ Determiner : supporting word.

* PENN'S Tree Bank

- PRP → Personal Pronoun I, you, he, she
- RB → Adverb → quickly, carefully over
- VBD → verb, past tense → ate, played
- DT → Determiner → The, A
- JJ → Adjective → quick, brown, wooden.
- NN → Single Noun → fox, vase, dog
- IN → Preposition → on, over, at, in, by, despite.
- VBG → verb, gerund or past participle
- * VBN → verb, non 3rd person singular present past participle
- * VBP → verb, 3rd person singular present → jumps, is
- VBZ → Third person singular present → jumps, is.
(verb with tense)
- VB → Normal verb.
- MD → Modal - can, will, should.
- WRB → adverb with question. how, where, whom.

① The quick brown fox jumps over a
 DT JJ JJ (feature of fox) NN VBZ IN DT
lazy dog
 JJ NN

② She carefully placed the glass vase on
 PRP RB VBD DT JJ NN IN
the wooden shelf
 DT JJ NN

③ I will call you when the meeting is
 PRP MD VB PRP WRB DT NN VBZ
over
 RB

④ Despite the rain they continued playing
 IN DT NN PRP VBD VBG
outside
 RB

VBG → Verb Gerund - eating playing

VBD → Verb past tense - continued eat

21/7/25

Encoder → Decoder

Levels of NLP

① Phonology / Phonetics

cat |k| lac |l|

② Morphology - study of structure of words
 Unhappiness → (un) + happy + ness

③ Lexical Analysis - to learn the grammatical structure of the sentence

Ram is a boy
 N V DT Noun

④ Syntactic Analysis - concentrate on the grammatical structure of phrase

⑤ Semantic Analysis - I am going to bank

⑥ Pragmatics : Can you pass the exam?
 - try to understand the intense meaning

Viterbi Algorithm

→ Dynamic Programming Algorithm.

The cat sleeps
DT NN VBZ

Finding the next pos tag, (Sequential strategy)
w.r.t. to the previous tag

Transition probability helps to identify the
next tag
(Heart of the algorithm).

Components

- 1) States → pos tags
- 2) Observations → Each word in the sentence
- 3) Transition probability → prob. Transition from one tag to another.
- 4) Emission probability (Observation Prob.)
 90% → sleeps → verb } so Major
 10% → sleeps adverb. } choose Verb.
- 5) start and end states.

EX: Corpus Training Data.

1) He reads books
PRP VBZ NN

2) She eats pizza
PRP VBZ NN

3) They enjoy football
PRP VBZ NN

4) Books are educational.
NN VBZ JJ (adjective)

5) Pizza is tasty.
NN VBZ JJ.

① Pos Tagging

NN - Noun	VB - verb	Modular - can, will, shall.
JJ - Adjective	VBZ - 's,	Determiner - A, An, The.
PRP - Pronoun. (He, she, we, they, I, you)	VBD - was	
IN - at, after.	VBG - ing	
	RB - Adverb	
	WRB - who/whom	
	VBN - many.	

② Viterbi Algorithm.

1) Initialization: $V[1][tag] = p[tag] * p[word/tag]$

(2) Iteration:

$$V[2][tag] = \text{Max} \left[V[1][\text{previous state}] \right. \\ \left. * p[tag | \text{previous state}] \right. \\ \left. * p[word / tag] \right]$$

① Pos Tagging

NN - Noun	VB - verb .	Modal - can, will, shall.
JJ - Adjective	VBZ - 's,	Determine
PRP - Pronoun. (He, she, we, they, I, you)	VBD - was	- A, An, The.
IN - at, after.	VBG - ing	
	RB - Adverb	
	WRB - who/whom	
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Q1 Corpus Training Data

- ① He | reads | books |
PRP VBZ NN
- ② She | eats | pizza |
PRP VBZ NN
- ③ They | enjoy | football |
PRP VBZ NN
- ④ Books | are | educational |
NN VBZ JJ
- ⑤ Pizza | is | tasty |
NN VBZ JJ

Test sentence: He eats football.

Initialization:

$$V[i] \text{ tag} = p[\text{tag}] * p[\text{word} / \text{tag}]$$

Postag	Start prob. $p[\text{tag}]$ (A)	Emission prob. $p[\text{word} / \text{tag}]$ ↓ He (B)	Viterbi Value = $A * B$.
PRP	$\frac{3}{5}$	$p[\text{He} / \text{PRP}] = \frac{1}{3}$	$\frac{1}{5} = 0.2 \checkmark$
NN	$\frac{2}{5}$	$p[\text{He} / \text{NN}] = 0$	0
VBZ	0	$p[\text{He} / \text{VBZ}] = 0$	0
JJ	0	$p[\text{He} / \text{JJ}] = 0$	0

Iteration

4.8
0.0048

$$(1) V[2][tag] = \max \{ V[1][\text{previous state}]$$

$$* P[tag | \text{previous state}]$$

$$* P[word | tag]$$

Post tag	Previous tag	tag / previous tag	word / tag	iteration value $A * B * C$
PRP	-	$P[- -]$	$P[\text{eats} / \text{tag}]$ $P[\text{eats} / \text{PRP}] = 0$	$0.2 * 0 = 0$
NN	VBZ	$P[\text{NN} / \text{VBZ}] = \frac{3}{5} = 0.6$	$P[\text{eats} / \text{NN}] = 0$	$0.2 * 0.6 * 0 = 0$
✓ VBZ	PRP	$P[\text{VBZ} / \text{PRP}] = \frac{3}{3} = 1$	$P[\text{eats} / \text{VBZ}] = \frac{1}{5}$	$0.2 * 1 * \frac{1}{5} = 0.04$
✓ VBZ	NN	$P[\text{VBZ} / \text{NN}] = \frac{2}{2} = 1$	$P[\text{eats} / \text{JJ}] = 0$	$0.2 * \frac{2}{5} * 0 = 0$
JJ	VBZ	$P[\text{JJ} / \text{VBZ}] = \frac{2}{5}$		

(2)

Post tag	Previous tag	$P(\text{tag} / \text{previous tag})$	Word / tag	iteration
PRP	-	-	$P[\text{Football} / \text{PRP}] = 0$	
✓ NN	VBZ	$P[\text{NN} / \text{VBZ}] = \frac{3}{5}$	$P[\text{Football} / \text{NN}] = \frac{1}{5}$	$\frac{3}{5} * \frac{1}{5} * 0.01$
VBZ	NN	$P[\text{VBZ} / \text{NN}] = 0$	$P[\text{Football} / \text{VBZ}] = 0$	
JJ	VBZ	$P[\text{JJ} / \text{VBZ}] = 0$	$P[\text{Football} / \text{JJ}] = 0$	
VBZ	PRP	$P[\text{VBZ} / \text{PRP}] = 0$	$P[\text{Football} / \text{VBZ}] = 0$	



Q2 Initial prob:

$$\pi(R) = 0.6 \quad \pi(S) = 0.4$$

Transition prob

$$a_{ij} = P(\text{next state} = j \mid \text{current state} = i):$$

$$a_{RR} = 0.7, \quad a_{RS} = 0.3, \quad a_{SR} = 0.4, \quad a_{SS} = 0.6$$

Emission prob $P(\text{rain} \mid \text{sunny})$

For Rainy: $b_R(\text{walk}) = 0.1, \quad b_R(\text{shop}) = 0.4, \quad b_R(\text{clean}) = 0.5$

For Sunny: $b_S(\text{walk}) = 0.6, \quad b_S(\text{shop}) = 0.3, \quad b_S(\text{clean}) = 0.1$

Given the observation sequence: [walk, shop, clean]
we use Viterbi algorithm to find the most likely state sequence and its probability.

Initialization.

Walk

Pos tag	$P(\text{tag})$	$P(\text{word} \mid \text{tag})$	Viterbi value $= P(\text{tag}) \times P(\text{word} \mid \text{tag})$
Rainy	0.6	$P(\text{walk} \mid \text{Rainy}) = 0.1$	0.06
<u>Sunny</u>	0.4	$P(\text{walk} \mid \text{sunny}) = 0.6$	$0.6 \times 0.4 = 0.24$

✓
Sunny

Iteration:

pos [tag]	p[previous tag]	p[tag/previous tag]	p[word/tag] p[walk/tag]	Viterbi value
Rainy	Rainy	0.7	0.1	$0.1 \times 0.7 \times 0.06$ 0.0042
Sunny	Sunny	0.3	0.1	$0.1 \times 0.3 \times 0.24$ 0.0072
Rainy	Rainy	0.4	0.6	$0.4 \times 0.6 \times 0.26$ 0.00624
Sunny	Sunny	0.6	0.6	$0.6 \times 0.6 \times 0.24$ 0.00864

Iteration : Shop

pos [tag]	p[previous tag]	p[tag/previous tag]	p[word/tag] p[shop/tag]	Viterbi value
Rainy	Rainy	0.7 RR	0.4	0.7×0.4 $\times 0.06$ 0.0168
Rainy	Sunny	0.4 RR	0.4	0.4×0.4 $\times 0.24$ 0.0384
Sunny	Rainy	0.3 RS	0.3	0.3×0.3 $\times 0.26$ 0.0234
Sunny	Sunny	0.6 SS	0.3	0.6×0.3 $\times 0.24$ 0.036

Rainy → Rainy.

Sunny → Sunny.

$$0.06 \times 0.7 \times 0.4 = 0.0168$$

$$0.24 \times 0.4 \times 0.4 = 0.0384$$

$$0.06 \times 0.3 \times 0.26 = 0.00468$$

$$0.24 \times 0.6 \times 0.3 = 0.0432$$

Viterbi value
Rainy = 0.0384

Viterbi value
Sunny = 0.0432



Iteration: clean.

Postay	Previous Day.	p[tag / previous]		Max / min word / tag clean / tag	Value
<u>Rainy</u>	Rainy 0.0384	0.7	arr	0.5	0.5×0.7 $\times 0.0384$
Rainy	Sunny 0.0432	0.4	ass	0.5	$0.5 \times$
Sunny	Rainy 0.0384	0.3	ars	0.1	$0.1 \times$
Sunny	Sunny 0.0432	0.6	ass	0.1	$0.1 \times$

Rainy $\checkmark 0.0384 \times 0.7 \times 0.5 = 0.01344$ ✓

$0.0432 \times 0.4 \times 0.5 = 0.00864$

$0.0384 \times 0.3 \times 0.1 = 0.01152$

$0.0432 \times 0.6 \times 0.1 = 0.002592$

$\underbrace{0.002592} \times 0.1$

Sunny
Rainy $\rightarrow$ Rainy $\rightarrow$ Rainy

$= 0.01344$

3/7/25

15 M

Very Important

Naive Bayes

Feature Extraction

convert text into numerical value

Text classification

X. one question CAT

I/p → Feature → ML model

Doc	Text	Class	No. of words
1	I loved the Movie	+ve ✓	4 ✓
2	I hated the Movie	-ve	4
3	A great movie, a good movie	+ve ✓	6 ✓
4	poor acting	-ve	2
5	Great Acting a good movie	+ve ✓	5 ✓

Perform a text classification using Naive Bayes Algorithm using the dataset Method (Bag of words)

Solution

(1) calculate the unique words (Vocabulary)

I loved the Movie hated great a good poor acting

Size of the Vocabulary = 10

(2) convert text into Numerical

2 Methods

1. TF → Term Frequency (Bag of words (BoS))

2. TF-IDF → Term Frequency - Inverse Document Frequency

Term Frequency (Bag of words) Method

Doc	I	loved	the	Movie	hated	great	a	good	poor	Acting
(1)	1	1	1	1	0	0	0	0	0	0
(2)	1	0	1	1	1	0	0	0	0	0
(3)	0	0	0	2	0	1	2	1	0	0
(4)	0	0	0	0	0	0	0	0	1	1
(5)	0	0	0	1	0	1	1	1	0	1

1114. $p(I/-) = 0.4$ $p(loved/-) = 0.125$ $p(hated/-) = 0.125$ $p(the/-) = 0.125$ $p(good/-) = 0.125$ $p(poor/-) = 0.125$ $p(acting/-) = 0.125$

ii) Probabilty of classes:

$$p(\text{classes}) \Rightarrow +, -ve.$$

= $\frac{\text{total No. of times particular class is present}}{\text{total No. of documents.}}$

total No. of documents.

$$p(+)=0.6=\frac{3}{5} \quad p(-)=0.4=\frac{2}{5}$$

no. of times word present in +ve class.

$$p(w_k|+) = \frac{n_{k+} + 1}{n_+ + |V|}$$

Smoothing function (Laplace)

$$p(w_k|-) = \frac{n_{k-} + 1}{n_- + |V|}$$

no. of times word present in -ve class.

total no. of words in a +ve / -ve class.

size of vocabulary.

Naive Bayes

$p(\text{classes}) = \frac{\text{total no. of times particular class is present}}{\text{total no. of documents}}$

total no. of

no. of words in positive statement = 15 = n_+

no. of words in negative statement = 6 = n_-

$$p(+) = \frac{3}{5}$$

$$p(-) = \frac{2}{5}$$

ex: n_{k+} = no. of I in positive statement.

$$p(I|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25} = 0.08$$

$$p(I|-) = \frac{(0) + 1}{(6) + 10} = \frac{1}{16}$$

$$p(loved|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25} = 0.08$$

$$p(loved|-) = \frac{(0) + 1}{(6) + 10} = \frac{1}{16}$$

$$p(hate|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25}$$

$$p(hate|-) = \frac{(1) + 1}{(6) + 10} = \frac{2}{16}$$

$$p(more|+) = \frac{(4) + 1}{(15) + 10} = \frac{5}{25}$$

$$p(more|-) = \frac{(1) + 1}{(6) + 10} = \frac{2}{16}$$

$$p(a|+) = \frac{(0) + 1}{(15) + 10} = \frac{1}{25}$$

$$p(a|-) = \frac{(0) + 1}{(6) + 10} = \frac{1}{16}$$

$$p(hated|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25}$$

$$p(hated|-) = \frac{(1) + 1}{(6) + 10} = \frac{2}{16}$$

$$p(great|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25}$$

$$p(great|-) = \frac{(0) + 1}{(6) + 10} = \frac{1}{16}$$

$$p(good|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25}$$

$$p(good|-) = \frac{(0) + 1}{(6) + 10} = \frac{1}{16}$$

$$p(acting|+) = \frac{(1) + 1}{(15) + 10} = \frac{2}{25}$$

$$p(acting|-) = \frac{(1) + 1}{(6) + 10} = \frac{2}{16}$$

$$p(poor|+) = \frac{(0) + 1}{(15) + 10} = \frac{1}{25}$$

$$p(poor|-) = \frac{(1) + 1}{(6) + 10} = \frac{2}{16}$$

$$p(+)=0.6$$

$$p(-)=0.4$$

Test sentence: I hated the poor acting.

$$p(+).p(I|+).p(hated|+).p(the|+).p(poor|+).p(acting|+)$$

Vector Space Model (VSM)

camlin

Date:

Text Classification

- Term Frequency Inverse Document Frequency
- TF IDF

→ convert text into numerical

→ Mainly use for feature extraction.

Doc	Text-	Class
1	I loved the movie	+
2	I hated the movie	-
3	A great movie, good movie	+
4	Poor Acting	-
5	Great Acting, a good movie.	+

1) Find the vocabulary

$V = I, \text{loved, the, movie, hated, A, great, good, poor, Acting}$

$|V| = 10$
→ term
→ document

TF-IDF (t, d)

$$= \text{TF}(t, d) * \text{IDF}(t, D)$$

$$= \frac{f_{t,d}}{N_d} * \log_2 \left(\frac{N}{n_t} \right)$$

(i) $\text{TF}(t, d) = \frac{\text{No. of times the term } t \text{ occurs in document } d}{\text{total No. of terms in document } d}$

$$= \frac{f_{t,d}}{N_d} \quad \left(\frac{\text{Frequency}}{\text{total No. of terms in document}} \right)$$

$$\text{IDF}(t, d) = \log \left(\frac{N}{n_t} \right) \quad \begin{array}{l} N \rightarrow \text{total no. of documents in no. word.} \\ n_t \rightarrow \text{No. of documents containing term.} \end{array}$$

frams must be 1 or 0 only Binary values

Term	term ₁	term ₂	term ₃	term ₄	term ₅	term ₆	term ₇	term ₈	term ₉	term ₁₀
I	1	1	0	0	0	1/4	1/4	0	0	0
loved	1	0	0	0	0	1/4	0	0	0	0
the	1	1	0	0	0	1/4	1/4	0	0	0
more	1	1	1	0	1	1/4	1/4	2/5	1	0
hated	0	1	0	0	0	0	1/4	0	0	0
a	0	0	1	0	1	0	0	1/5	0	0
great	0	0	1	0	1	0	0	1/5	0	0
good	0	0	1	0	1	0	0	1/5	0	0
poor	0	0	0	1	0	0	0	0	0	1/2
Acting	0	0	0	1	0	0	0	0	0	1/2

N → No. of sentences = 5

Term	n_t	$\log_2(N/n_t)$	$TF \times IDF_1$	$TF \times IDF_2$	$TF \times IDF_3$	$TF \times IDF_4$
I	2	$\log_2(5/2) = 1.32$	$0.25 \times 1.32 = 0.33$	$0.25 \times 1.32 = 0.33$	0	0
loved	1	$\log_2(5/1) = 2.32$	$0.25 \times 2.32 = 0.58$	0	0	0
the	2	$\log_2(5/2) = 1.32$	$0.25 \times 1.32 = 0.33$	$0.25 \times 1.32 = 0.33$	0	0
more	4	$\log_2(5/4) = 0.32$	$0.25 \times 0.32 = 0.08$	$0.25 \times 0.32 = 0.08$	$0.4 \times 0.32 = 0.128$	0
hated	1	$\log_2(5/1) = 2.32$	0	$0.25 \times 2.32 = 0.58$	0	0
a	2	$\log_2(5/2) = 1.32$	0	0	0	0
great	2	$\log_2(5/2) = 1.32$	0	0	0	0
good	2	$\log_2(5/2) = 1.32$	0	0	0	0
poor	1	$\log_2(5/1) = 2.32$	0	0	0	0
Acting	2	$\log_2(5/2) = 1.32$	0	0	0	0

$$p(\text{word} | \text{class}) = \frac{n_{k+} + 1}{n + |V|}$$

$$(ii) p(\text{word} | +) = \frac{n_{k+} + 1}{n + |V|}, p(\text{word} | -) = \frac{n_{k-} + 1}{n + |V|}$$

$$(i) p(I | +) = \frac{0.33 + 0 + 0 + 0 + 0}{0.33 + 0 + 0 + 0 + 0 + 1}$$

$$p(I|+) =$$

$$p(\text{love}|+) =$$

$$p(\text{movie}|+) =$$

$$p(\text{hated}|+) =$$

$$p(a|+) =$$

$$p(\text{great}|+) =$$

$$p(\text{good}|+) =$$

$$p(\text{poor}|+) =$$

$$p(\text{acting}|+) =$$

$$p(I|-) =$$

$$p(\text{loved}) =$$

$$p(\text{more}) =$$

$$p(\text{hated}) =$$

$$p(a|-) =$$

$$p(\text{great}) =$$

$$p(\text{good}) =$$

$$p(\text{poor}) =$$

$$p(\text{acting}) =$$

Test sentence:

I hated the movie.

$$+ p(+) * p(I|+) * p(\text{hated}|+) * p(\text{the}|+)$$

$$\frac{3}{5} * \frac{1-33}{24} * p(\text{movie}|+)$$

$$- p(-) * p(I|-) * p(\text{hated}) * p(\text{the})$$

$$* p(\text{Movie})$$

$$\frac{2}{5}$$